a) Agent-driven Hierarchical Evidence Bank Construction

Global Survey

Input WSI pyramid
$S = \{L_0, \ldots, L_H\}$

Morphometric

Tumor size
2.8 mm
Principal axis
11.02mm

tumour
necrosis
adipose
stroma
other

G_s global record
43 tiles, tissues 40.1%, carcinoma-like 76.7%

Coarse-to-fine agentic sampling

Morphology Agent
Tissue Segment | Clinical Codebook | Pathology VLM

Tile triage ranking
Tile.12: tumour 0.91
Tile.07: stroma 0.62
Tile.23: necrosis 0.55

Hierarchical Visual Evidence
Tiles → Anchors A_s → Patches P_s

Selected views
tissues | 10x morphology | 20x cellular

Visual-textual observation

Pathology VLM
Quilt-LLaVA, Patho-R1, etc

Densely packed cells, enlarged irregular nuclei, Mitotic figures present
conf 0.86 kept

Solid architecture with duct-like lumina; irregular tumour-stroma interface
conf 0.79 kept

Out-of-focus/ folded tissue, not assessable
conf 0.34 discarded

$D_a = \{d_p | p \in P_a\}$
Linked to patch ID and L_0 coordinates

Pathology signal grounding

Convert free-form text to predefined concepts

Clinical codebook K

nuclear pleomorphism	marked
mitotic activity	high
invasive interface	present
tubule / gland formation	poor
necrosis	focal
stromal reaction	desmoplastic
lymphocytic infiltrate	absent

Signal record σ

Signal k	invasive interface
Status y	present
Support	p3, p7, anchor a2
Location	(18432, 9216) at 20x

Hierarchical Evidence Bank M_s

Signal S_s: pleomorph. | invasion | necrosis
Global view G_s
Anchors A_s: a_1 a_2 a_3
Patches P_s: p_1 p_2 p_3 p_4 p_5 p_6

signal	status	supp.	L0 loc. @ mag
invasive interface	present	p3,p7	(18432, 9216) @20×
nuclear pleomorph.	marked	p1,p4	(20096, 7680) @20×
necrosis	focal	p6	(15104, 11520) @10×
tubule formation	poor	p2,p5	(19840, 8704) @20×

$M_s = \{d_p | p \in P_a\}$
signal→description→images→coordinate

b) Hierarchical Evidence-Augmented WSI Reasoning

User question q_t
M_s
Question-to-concept routing

G-level
tissue 40.1% · carcinoma 76.7% → "no" $\gamma = 0.61$

A-level
irregular tumour-stroma border → "yes" $\gamma = 0.88$
desmoplastic stromal reaction → "yes" $\gamma = 0.84$

P-level
cells infiltrating the stroma → "yes" $\gamma = 0.91$
pleomorphic nuclei, mitoses → "yes" $\gamma = 0.86$
duct-like lumina, necrosis → "no" $\gamma = 0.55$

Top-k Selection
weighted answer consensus
answer

c) Persistent Evidence Reuse Across Queries

M_s

Turn 1 q_1: "Signs of invasion"
σ: invasive interface → G

Turn 2 q_2: Details of Tumor type
σ: pleomorphism, tubule → $G + A + P$

Turn 29 q_{29}: "molecular subtype?"
σ: pleomorphism → $G + A$

94.3 s → 5.9 s

ΔAcc	Rephrase consistency	online cost
-1.38	99.47%	2.73s
v.s. single turn	5 paraphrasing	Per query